

MANISH KUMAR TIWARI

SOFTWARE ENGINEER | AI & BACKEND SYSTEMS | DISTRIBUTED SYSTEMS

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LeetCode

PROFESSIONAL SUMMARY

Software Engineer specializing in backend systems, AI/LLM applications, distributed processing, and production software architecture. Experienced in building **asynchronous execution systems**, **LLM-powered structured workflows**, **automated evaluation pipelines**, **RAG systems**, and **sandboxed compute infrastructure** using Python, Java, TypeScript, Node.js, FastAPI, PostgreSQL, Redis, and Docker. Production experience developing software at **Beans.ai**, with a strong foundation in data structures, algorithms, object-oriented design, databases, and systems engineering.

TECHNICAL & PROFESSIONAL SKILLS

AI & GenAI:	LLMs, Generative AI, Prompt Engineering, Structured Outputs, Function Calling, Embeddings, RAG, Gemini API, NLP, Conversational AI
AI Evaluation:	Output Validation, Deterministic Evaluation, Classification, Information Extraction, Error Analysis, Edge-Case Testing, Schema Validation, Confidence Scoring
Backend & Systems:	Python, FastAPI, Node.js, Express.js, Spring Boot, REST APIs, Redis, BullMQ, Socket.IO, Asynchronous Processing, Worker Architecture
Programming:	Python, Java, JavaScript, TypeScript, Swift
Data & Databases:	PostgreSQL, MongoDB, MySQL, Prisma, SQL, Database Design
DevOps & Security:	Docker, Docker Compose, Linux, Git, GitHub, Postman, CI/CD, Process Sandboxing, Network Isolation, Resource Limits
Engineering:	OOP, Clean Architecture, MVVM, Dependency Injection, Modular Design, Concurrency, Automated Testing, API Integration

PROFESSIONAL EXPERIENCE

Beans.ai Noida, India | May 2024 – Present
Software Engineer — iOS & Production Systems PPO Conversion

- Production Application Architecture:** Rebuilt approximately **80% of the Beans Route iOS application** using SwiftUI, MVVM, and Clean Architecture, restructuring core application flows into maintainable and reusable modules.
- Networking & Concurrency:** Implemented asynchronous networking, dependency injection, REST API integrations, Swift concurrency, and reusable UI components across production application workflows.
- Modular Architecture:** Reduced third-party dependencies by replacing reusable functionality with internal components and abstractions, improving maintainability and architectural control.
- Development Automation:** Automated repetitive development and code-generation workflows, reducing boilerplate and accelerating implementation of recurring engineering patterns.
- Production Engineering:** Worked within an existing production codebase, integrating new functionality while preserving established application behavior and backend API contracts; continued after internship conversion through PPO.

KEY ENGINEERING, AI & DATA SCIENCE PROJECTS

CampusAgent — Autonomous AI Agent Runtime & Evaluation Engine Python 3.12 • FastAPI • PostgreSQL • Docker • SSE • LLM Function Calling GitHub

- Agent Runtime:** Built an autonomous agent runtime supporting **multi-step reasoning**, **dynamic tool selection**, **structured function calling**, **working memory**, and **self-reflection loops**.
- Dynamic Tool Calling:** Designed schema-driven tool interfaces using **JSON Schema**, allowing agents to dynamically select and invoke tools while maintaining structured execution state.
- Evaluation Harness:** Developed deterministic evaluation workflows measuring **tool-selection accuracy**, **execution efficiency**, and **output-format compliance** across test scenarios.
- Reliability & Streaming:** Implemented LLM fallback handling, token-budget controls, rate limiting, automated backend testing, and real-time **SSE execution streaming**.

CampusBrain — Multi-Tenant Enterprise Hybrid RAG Platform FastAPI • Qdrant • BGE-M3 • Docker Compose • React GitHub

- Hybrid Retrieval:** Built a retrieval system combining **dense vector search with BM25 keyword retrieval through Reciprocal Rank Fusion (RRF)** for improved recall.
- Universal Document ETL:** Developed ingestion workflows supporting multiple document formats with **chunking**, **embeddings**, **metadata handling**, and **citation-aware retrieval**.
- Multi-Tenant Isolation:** Implemented organization-level vector-store isolation to maintain independent knowledge bases and prevent cross-tenant retrieval.
- RAG Architecture:** Integrated **BGE-M3 embeddings**, **Qdrant vector search**, and **reranking** into an end-to-end retrieval pipeline for grounded LLM applications.

NEXTHIRE — Technical Interview & Code Evaluation Platform React • TypeScript • Node.js • PostgreSQL • Prisma • Redis • BullMQ • Docker GitHub

- Asynchronous Judge Architecture:** Designed a Redis/BullMQ pipeline with dedicated workers separating API requests from compute-intensive code evaluation for Python, C++, and Java submissions.
- Secure Code Sandboxing:** Isolated untrusted programs using Docker with **network isolation**, **read-only filesystems**, **dropped capabilities**, **non-root execution**, **resource limits**, and **wall-clock timeouts**.
- Deterministic Evaluation:** Implemented explicit verdicts for **compilation**, **correctness**, **runtime**, **timeout**, **memory-limit**, and **output-limit failures**.
- Automated Testing:** Backed critical server and evaluation paths with **219 passing automated tests** and implemented real-time execution updates using Socket.IO.

GitaBot — Quantized Semantic Search & RAG Engine Python • FAISS • Mistral Embeddings • RAG GitHub

- Semantic Retrieval:** Built a semantic QA engine using **Mistral embeddings and FAISS-based vector retrieval** for context-grounded question answering.
- Memory Optimization:** Applied **8-bit scalar quantization** to reduce vector-index memory requirements while maintaining practical retrieval performance.
- Resource-Constrained Deployment:** Optimized deployment for a constrained **512 MB RAM environment**, targeting low-latency semantic retrieval.

EDUCATION, HONORS & PROFESSIONAL CERTIFICATIONS

Sitare University (with SRMU) Lucknow, India | GPA: 9.46/10 | 2024 – 2027
Bachelor of Technology (B.Tech) in Computer Science Expected Jul 2027

- Honors & Scholarships:** Awarded **100% B.Tech CS Scholarship** by Sitare Foundation (Oct 2024 – Present); Yuva Unstoppable Foundation Scholarship (Jul 2021 – Present).
- Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Artificial Intelligence, Machine Learning, Python Programming, Java Programming.